WW-C-581E
April 6, 1977
SUPERSEDING
Fed. Spec. WW-C-581d
June 5, 1962

FEDERAL SPECIFICATION

CONDUIT, METAL, RIGID, AND INTERMEDIATE; AND COUPLING, ELBOW, AND NIPPLE, ELECTRICAL CONDUIT: STEEL, ZINC COATED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers steel electrical conduits, conduit couplings, elbows, and nipples of the following types: rigid metal and intermediate metal. Exterior surfaces are zinc coated, and interior surfaces are coated with either zinc, enamel, or other equivalent corrosion-resistant coatings.

1.2 Classification. Conduit, couplings, elbows, and nipples furnished under the specification shall be one of the following classes, sizes, types, and styles as specified (see 6.2). Sizes are nominal or trade sizes in inches. Nipple lengths shall be standard commercial length as specified (see 6.2).

Class 1 - Rigid metal steel conduit, couplings, elbows, and nipples.

Sizes: 1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4, 5, and 6.

Class 2 - Intermediate metal steel conduit (IMC), couplings, elbows, and nipples.

Sizes: 1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, and 4.

Type A - Conduit.
Type B - Coupling.
Type C - Elbow.

Style 1 - Standard 90° elbow.
Style 2 - Large radius 90° elbow.
Style 3 - Standard 45° elbow.

Type D - Nipple.

FSC 5975

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-116 - Preservation-Packaging, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-163 - Steel Mill Products Preparation for Shipment and Storage.

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(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute, Inc., (ANSI):

C80.1 - Rigid Steel Conduit, Zinc-Coated.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Underwriters' Laboratories, Inc. (UL):

UL6 - Rigid Metallic Conduit.
UL1242 - Intermediate Metal Conduit.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 General requirements. Conduit, couplings, elbows, and nipples furnished under this specification shall conform as applicable, to ANSI C80.1, UL6, or UL1242, and shall be made of steel. Unless otherwise

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specified (see 6.2), IMC may be either UL type I IMC or UL type II IMC as specified in UL1242. Each 10-foot length of conduit shall be furnished with one coupling attached.

3.2 Standard commercial product. The conduit, couplings, elbows, and nipples shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements. Additional or better features which are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product shall be included in the conduit, couplings, elbows, and nipples being furnished. Standard commercial product is a product which has been or will be sold on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model(s).

3.3 Identical items. Conduit, couplings, elbows, and nipples of the same classification furnished under any specific contract shall be physically and mechanically identical. No deviation will be acceptable without prior written approval of the contracting officer.

3.4 Material. All material shall be new and unused. All material shall be as specified herein. Material not specified shall be of the same quality used for the intended purpose in commercial practice. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual component or the overall assembly.

3.5 Fire and casualty hazards.

3.5.1 Each contractor shall submit, to the contracting agency, proof that the conduit, couplings, elbows, and nipples he proposes to supply under this specification conforms to the requirements of UL6 or UL1242 as applicable. The UL listing mark may be accepted as evidence that the conduit, couplings, elbows, and nipples conform to this requirement.

3.5.2 In lieu of the UL listing mark, the contractor may submit independent proof satisfactory to the contracting agency, that his conduit, couplings, elbows, and nipples conform to the applicable requirements of UL6 or UL1242, including methods of tests.

3.5.3 Compliance with the above preliminary requirements in regards to fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure acceptance of his materials.

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3.6 Marking. Conduit, elbows, and nipples shall be permanently marked in accordance with ANSI C80.1, UL6, or UL1242, as applicable.

3.7 Workmanship. All threaded portions of conduit, couplings, elbows, and nipples shall be clean and undamaged. Plastic thread protectors shall be furnished on all exposed conduit threads. The exterior zinc coating, and all other protective coatings, including the interior coating and thread coating, shall completely and uniformly cover the metal substrate.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Preparation for delivery inspection (see 4.7).

4.3 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.5 and the tests of 4.6.

4.3.1 Inspection lot. A lot shall consist of all conduit, couplings, elbows, and nipples of the same classification offered for delivery to the Government at one time under a specific contract.

4.4 Sampling. A random sample of conduit, couplings, elbows, and nipples shall be selected from each lot in accordance with MIL-STD-105.

4.4.1 Sampling for examination. Examination of the conduit, couplings, elbows, and nipples shall be based on inspection level S-3 and an Acceptable Quality Level (AQL) of 4.0 percent defective.

4.4.2 Sampling for tests. Tests of the conduit, couplings, elbows, and nipples shall be based on inspection level S-3 and an AQL of 4.0 percent defective.

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4.5 Examination. Each sample selected in accordance with 4.4.1 shall be examined for compliance with the requirements specified in section 3 of this specification and UL6, ANSI C80.1, or UL1242, as applicable. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examination and dimensional measurements. The UL listing mark may be accepted as evidence that the conduit, couplings, elbows, and nipples conform to UL6 or UL1242, as applicable.

4.6 Tests. Each sample selected in accordance with 4.4.2 shall be tested as specified in ANSI C80.1, UL6, or UL1242, as applicable. Class 2 conduit, couplings, elbows, and nipples shall also be tested as specified in ANSI C80.1. The UL listing mark may be accepted as evidence that the conduit, couplings, elbows, and nipples meet the tests of the applicable UL Standards. Failure of the conduit, couplings, elbows, and nipples to meet the requirements of the tests shall constitute cause for rejection.

4.7 Preparation for delivery inspection. The packaging, packing, and marking of the conduit, couplings, elbows, and nipples shall be inspected to verify conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Conduit. The conduit shall be packaged in accordance with MIL-STD-163, level A.

5.1.1.2 Couplings, elbows, and nipples. The couplings, elbows, and nipples shall be preserved and packaged in accordance with MIL-P-116, method III.

5.1.2 Level C. The conduit, couplings, elbows, and nipples shall be preserved and packaged in a manner that will insure adequate protection against deterioration and damage during shipment. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

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5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Conduit. The conduit shall be packed in accordance with MIL-STD-163, level A.

5.2.1.2 Couplings, elbows, and nipples. The couplings, elbows, and nipples of like description shall be packed in close-fitting boxes conforming to PPP-B-621, class 2; or PPP-B-601, overseas type. The contents shall be blocked and braced to prevent movement or damage.

5.2.2 Level B.

5.2.2.1 Conduit. The conduit shall be packed as specified for level A.

5.2.2.2 Couplings, elbows, and nipples. The couplings, elbows, and nipples shall be packed as specified for level A except that the boxes shall be domestic type, class I; or in boxes conforming to PPP-B-636.

5.2.3 Level C. The conduit, couplings, elbows, and nipples shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules, or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or order, all marking shall be in accordance with 5.3.1 or 5.3.2 as specified (see 6.2).

5.3.1 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The conduit, couplings, elbows, and nipples are intended for use as wire and cable raceways in an electrical power distribution system.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class, size, type, and style (if applicable) of conduit, couplings, elbows, and nipples required (see 1.2).
(c) Length of nipple required (see 1.2).
(d) When UL type I IMC or UL type II IMC is required (see 3.1).
(e) Level of preservation and packaging, and level of packing required (see 5.1 and 5.2).
(f) Marking required (see 5.3).

6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Armed Services Procurement Regulation (ASPR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 It is believed that this specification adequately describes the characteristics necessary to secure the desired conduit, couplings, elbows, and nipples, and that normally no samples will be necessary prior to award, to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid samples should be definitely stated, the specification to apply in all other respects.

6.5 Cross reference of classifications. Revision E of this specification adds class, size, type, and style classifications (see 1.2). The title of the specification has been changed to include intermediate metal steel conduit.

Previous classification WW-C-581d	New classification WW-C-581E
Size (as specified)	Class 1 Size (as specified) Type A

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MILITARY INTERESTS:

Custodians:

Army - ME
Air Force - 85

Review activity:

Air Force - 99

User activities:

Army - MI
DSA - GS

Civil Agency Coordinating Activities:

GSA - FSS, PBO
NASA - JFK
USDA - AFS

Preparing activity:

Navy - YD

Project No. 5975-0528

Orders for this publication are to be placed with General Services Administration, acting as agent for the Superintendent of Documents. See section 2 of this specification to obtain copies of documents referenced herein. Price 35 cents each.

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